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Impacts of sterilization method on material properties of 3D printable resins with prolonged exposure to cell culture environments

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Abstract

Low-cost 3D printing has become increasingly important in biomedical research, enabling rapid fabrication of custom cell culture devices, fixtures, and biohybrid robotic components. However, little is known about how common sterilization procedures and prolonged cell culture exposure impact the mechanical properties of commercially available resins. In this study, we analyzed five candidate 3D-printable resins—three rigid (Asiga DentaGUIDE, Liqcreate Bio-Med Clear, Phrozen AquaGray 8K) and two elastomeric (Asiga DentaGUM, Formlabs Silicone 40A)—to evaluate the effects of ethanol/UV and autoclave sterilization on material properties in phosphate-buffered saline (PBS) at physiological conditions. Using stress-strain data from tensile and compressive mechanical tests, elastic moduli, ultimate tensile strength, and strain at break were compared across sterilization techniques. Results showed that sterilization significantly altered the mechanical properties of rigid resins, with Phrozen AquaGray 8K and Liqcreate Bio-Med Clear exhibiting large reductions in tensile stiffness and strength, while Asiga DentaGUIDE retained greater stability. In contrast, elastomeric resins were more robust: Asiga DentaGUM and Formlabs Silicone 40A demonstrated minimal or non-significant changes across sterilization methods, though post-treatment protocols influenced variability. Notably, several rigid resins also exhibited substantially lower moduli in PBS compared to manufacturer-reported values. These findings emphasize the need for experimental validation of 3D printed resin properties under intended use conditions and suggest that Asiga DentaGUIDE and Formlabs Silicone 40A are promising candidates for biohybrid applications. Overall, this work provides critical guidance for selecting sterilization-compatible materials for biomedical devices and biohybrid robotics.

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1 Introduction

In biomedical research, 3D printers have become ubiquitous. The ability to rapidly prototype and deploy custom devices in the lab enables higher throughput of device prototyping and makes custom components more accessible. Furthermore, with the rapid advancement of additive manufacturing techniques, several biocompatible, 3D-printable materials have become readily and commercially available, and the resolution of printed features has greatly improved. In particular, resin 3D printers can currently achieve resolutions at the 10s of microns scale at the cost of $1000s (USD) or less [1], increasing the accessibility to high-quality prints at low cost. Example systems as of October 2025 include Asiga's MAX X27 (27 µm resolution, $9,990 [2], [3]), Formlabs' Form 4B (50 µm XY resolution, $5,970 [4]), and Phrozen's Sonic Mini 8K (22 µm resolution, $398.99 [5]). The reduced cost of 3D printing and the increase in the available biocompatible materials increases the applicability and accessibility of custom parts for biomedical applications. Whether designing custom cell culture chambers, sample fixtures, novel devices, or biohybrid robots, 3D printing custom parts for prototypes and even deployment is becoming increasingly common.

For devices and components that must withstand or respond to mechanical loads, having robust models of material properties is critical for design. This is particularly important in the growing field of muscle-based biohybrid robotics, where researchers seek to integrate living muscle and abiotic components to build functional devices [6], [7], [8]. In such systems, the muscle actuators contract and apply loads to an engineered skeleton, such as thin films [9], [10], [11], [12] and parallel posts [13], [14], [15], [16], [17], which the latter would be akin to single-stroke actuation, though multiple cultured antagonistic muscle strips could be combined for double-stroke-like behavior [18]. These current biohybrid actuators have been developed to produce small forces on the µN to mN scale (or kPa scale when accounting for size) (for a review, refer to [6]). In all cases, the mechanics of the skeleton are critical to the design of a biohybrid actuator since muscle is mechanosensitive.

Research has demonstrated that environmental stiffness can have a significant impact on the differentiation and maturation of functional muscle tissue [19], [20]. Myotube striation was shown to be affected by the flexibility of the culture substrate [21]. If the substrate stiffness deviates from the stiffness of normal muscle (passive Young's modulus of $\approx$ 12 kPa), myotubes fail to form

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properly striated fibers [21]. In biohybrid actuators, the stiffness of the skeleton can impact the force generation over time [13], [14], [17]. As more 3D printable commercial resins are being shown to be biocompatible with proper post-processing [1], [22], this provides a scalable manufacturing approach for building skeletons for biohybrid robots, such as stiff structural supports [23] or flexible skeletons capable of motion [15].

Since muscle is sensitive to the stiffness of the environment, the mechanical properties of any interfacing, synthetic materials in biohybrid actuators must be well characterized. 3D printable materials have already been demonstrated to be sensitive to processing needed for use with cell culture. Notably, 3D printable materials can be impacted by sterilization [24], [25], [26]. However, a majority of commercial resin manufacturers do not report post-sterilization mechanical properties. Furthermore, prolonged exposure to typical cell conditions, such as warm (37°C) conditions submerged in saline, may impact the mechanical properties of 3D printed parts. In particular, polylactic acid (PLA), although more commonly used as a filament instead of as a resin in 3D printing, has demonstrated changes to prolonged water immersion at 37°C and 70°C in addition to differing properties when tested under submerged conditions [27], [28]. Additionally, 16 hours of immersion in boiling water has been used as a hydrothermal method to study accelerated aging of dental resins (VarseoSmile Crown plus and VarseoSmile Temp), which resulted in changes to flexural strength [29]. If post-processing sterilization techniques or prolonged exposure to cell culture conditions alter the mechanical properties of these 3D printable materials, the expected maturation and force development of muscle cells in contact with these materials could be impacted. Therefore, for use of these 3D printable materials in contact with cells, particularly mechanosensitive cells like muscle, the mechanical properties must be characterized after sterilization and under cell culture conditions. To the best of our knowledge, the mechanical properties of the resins in this study have also not been reported under typical cell culture conditions.

In this paper, we analyze mechanical testing data from biocompatible sterilized samples printed using commercial resins [1]. In particular, we sought to investigate two research questions: (1) How do standard sterilization techniques impact the mechanical properties of 3D printable resins? and (2) How does prolonged exposure to cell-culture-like environments alter the mechanical properties

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compared to manufacturer-reported values? For completeness, we provide a brief overview of the methods used to generate the published dataset. We then conduct statistical analyses to determine whether the mechanical properties are affected by a combined ethanol and UV sterilization or autoclave sterilization. Findings are reported for several commercially available elastic and rigid resins. As a result of this analysis, we highlight the importance of verifying the mechanical properties of post-sterilized samples in the intended usage conditions, as many samples were found to be significantly affected by sterilization. Additionally, for rigid resins, even nonsterilized samples exhibited lower moduli than those reported by the manufacturer when tested in cell culture environments.

2 Methods

In this paper, changes in the mechanical properties of five commercially available 3D-printable resins due to two sterilization techniques are investigated (Fig. 1). The changes are specifically investigated in the context of samples that are exposed to prolonged mammalian cell culturing conditions. The data analyzed in this paper have been previously reported in a data descriptor as part of a larger repository of material properties, cytotoxicity, and print fidelity data for these resins [1]. For a detailed description of the methods, analysis, and validation, we refer the reader to that data descriptor. Here, we briefly summarize the experiments and methods relevant to the particular analysis conducted here.

2.1 Sample Fabrication and Post-Processing

Five commercial resins (three rigid and two elastomeric) were assessed for their potential use with C2C12 muscle-based biohybrid actuators: (Rigid) Asiga DentaGUIDE, Liqcreate Bio-Med Clear, Phrozen AquaGray 8K; (Elastomeric) Asiga DentaGUM, Formlabs Silicone 40A. For mechanical tests, dogbone samples (for tensile testing) and cylinders (for compression testing) were printed using the Phrozen Sonic Mini 8K 3D printer (Phrozen Tech Co., LTD., Hsinchu, Taiwan). Part files, resin printing profiles, and print settings can be found in the accompanying data repository [1]. Printed parts of all resins were subjected to the same post-printing protocol consisting of two 5-minute sonicated rinses in 100% isopropanol (IPA), a 30-minute drying period, and two

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15-minute post-curing periods. For the Formlabs Silicone 40A, the manufacturers also had a proprietary post-printing wash solution (80% IPA, 20% *n*-butyl acetate, referred to in the rest of the paper as IPA/BuOAc). Therefore, a second set of samples was also printed and subjected to the manufacturer's recommended protocol. After post-processing the samples, the resins were sterilized using one of two sterilization methods (ethanol submersion or autoclaving) or left unsterilized as a control. Prior to mechanical testing, all samples were incubated in 1x PBS at 37°C for 16 hours to mimic mammalian cell culture conditions. For sample sizes and post-processed average geometries just prior to mechanical testing, see Tables 1 and 2.

2.2 Mechanical Testing

Uniaxial tension and compression tests were conducted using an MTS Criterion Model 42 (C42) universal test stand according to ASTM Standards D638-14 (rigid tensile [30]) and D412-06a (elastomeric tensile [31]), and D695-23 (rigid compression [32]) and D575-91 (elastomeric compression [33]), respectively. For complete protocol details, see [1]. All tests were conducted using a 50-N max capacity load cell with samples submerged in 37°C 1x PBS using an MTS BioBath. Time, force, and displacement data were recorded at 10 Hz for analysis. Prior to testing, the rest geometry (cross-sectional dimensions and gauge length) was measured for each sample.

2.3 Data Analysis and Constitutive Modeling

Using force and extension data from the MTS and rest configuration data measured prior to testing, engineering stress (force / rest cross-sectional area) and strain (displacement / gauge length) were calculated for each specimen. All stress/strain data were resampled by linear interpolation at 50 equally spaced strain points.

For rigid resins, the tensile and compressive moduli were estimated as the maximum instantaneous slope of the tension and compression stress-strain curves, respectively, according to ASTM standard D638-14 [30]. The line passing through the point of greatest instantaneous slope was also used to compensate for potential toe-region data. For the elastomeric data, the stress-strain data were fit to a 3-parameter Yeoh hyperelastic model [34] as this model can flexibly capture the chang-

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ing stiffness of material and be fully calibrated from only uniaxial data. Here, the toe region was accounted for by ignoring data below 5% of the maximum force achieved and assuming a linear region at low strains [1]. The hyperelastic model was fit to tensile and compressive data simultaneously using a bootstrapping approach [1]. For the Yeoh model, the Young's modulus (the tangent slope of the constitutive model at zero strain) was calculated from the model parameters. In addition to the constitutive model parameters, several scalar metrics were also calculated. From tensile tests, the Ultimate Tensile Strength (UTS) and Strain at Break were measured as the maximum stress and strain, respectively, that the specimen experienced. Note that this may not occur at the sample point for samples that soften before rupture. For the elastomer samples, only the DentaGUM broke within the machine limits, and therefore, these metrics were not recorded for the Formlabs Silicone samples. For elastomeric resins, the stress at 50% and 100% strain were calculated to compare to manufacturers' metrics.

2.4 Statistical Analysis

The mechanical properties extracted based on the raw data (all metrics except elastomer model parameters) were statistically analyzed using Minitab (v. 22.1; Minitab, LLC., State College, PA, USA). The Anderson-Darling test and the Bartlett test were used to assess normality and equivalent variances, respectively. Based on these initial tests, a majority of the groups were normal and did not have significantly different variances. Therefore, a General Linear Model (ANOVA) was fit to the data with a post-hoc Bonferroni pairwise comparison. A level of significance of $\alpha = 0.05$ was used for all analyses.

The Young's moduli of the elastomer resins were calculated from the bootstrapped model parameters of the Yeoh model (see 2.3), and therefore, we do not have modulus values for individual samples as would be needed for standard ANOVA analysis. Instead, inter-group comparisons were conducted non-parametrically using confidence intervals. For a given comparison, all pairwise differences between the bootstrapped estimates of the Young's moduli were calculated to obtain a bootstrap distribution of the group differences. The 95% ($100\times(1\text{-}\alpha)$) confidence interval of this comparison was determined from this distribution. If this confidence interval did not contain 0,

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then the groups are significantly different at the 5% level. If the interval contained 0, the groups are not statistically different. This procedure is analogous to the t-tests conducted in the post-hoc Bonferroni comparisons. This analysis was conducted using MATLAB R2024a (Mathworks, Natick, MA, USA). For results comparing elastomeric Young's moduli, the confidence interval is reported instead of a p-value.

3 Results

Using the data reported in Liao *et al.* 2025 [1], we analyzed the effects that autoclave and ethanol/UV sterilization had on the mechanical properties of commercially available rigid and elastomeric resins. All percent differences reported below are relative to the respective nonsterile group unless otherwise stated. For the magnitude of the different resin properties, see Fig. 2 (rigid) and Fig. 4 (elastomers). For constitutive models of the different materials, see Fig. 3 (rigid) and Fig. 5 (soft).

3.1 Sterilization causes changes in tested rigid resin properties

The data descriptor and associated raw data previously published by Liao *et al.* [1], included mechanical testing data for three commercially available rigid resin samples with three sterilization conditions. Using the mechanical testing data reported, we investigated the effect of sterilization technique on Young's modulus, Ultimate Tensile Strength (UTS), and elongation at break from the tensile tests, and the compressive modulus from the compression tests (see Fig. 2 for all rigid resin data comparisons).

3.1.1 Asiga DentaGUIDE

Compared to the other rigid resins tested (see below), Asiga DentaGUIDE performed the best under the different sterilization techniques, only seeing a significant increase in the compressive modulus for both groups (Fig. 2 Column 1; autoclave: $15.1 \pm 0.9\%$, ethanol/UV: $17.7 \pm 0.8\%$). Both groups also saw similar increases in their elongation at break relative to the nonsterile group (autoclave: 24 ± 8 higher than nonsterile, ethanol/UV: $19 \pm 9\%$ higher), though for neither group was this

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increase significant. The two sterilization groups differ in the change that was seen in the Young's moduli and in the Ultimate Tensile Strength. The autoclaved samples saw a $10 \pm 2\%$ decrease in Young's modulus and a $4.1 \pm 0.6\%$ increase in UTS relative to the nonsterile group, while the ethanol/UV group saw a $4.1 \pm 0.7\%$ increase in Young's modulus and $16 \pm 4\%$ decrease in UTS. However, these differences were not significant compared to the nonsterile group. Overall, both groups performed well under sterilization, with the autoclave group becoming slightly softer but with better failure characteristics.

3.1.2 Liqcreate Bio-Med Clear

Sterilization had a large impact on the mechanical properties of Liqcreate Bio-Med Clear. Both sterilization techniques saw statistically significant increases in sample compressive moduli (autoclave: $14.0 \pm 1.0\%$ increase relative to nonsterile, ethanol/UV: $20 \pm 2\%$ increase) and decreases in Ultimate Tensile Strengths (autoclave: $18 \pm 3\%$ decrease relative to nonsterile, ethanol/UV: $21 \pm 3\%$ decrease) (Fig. 2 Column 2). In addition, the autoclaved group saw a statistically significant decrease in Young's modulus ($29 \pm 4\%$ relative to nonsterile). The ethanol/UV group also saw a large decrease in Young's modulus ($24 \pm 7\%$), but the variance in this group means this difference did not reach significance. Similarly, both groups saw nonsignificant increases in their elongation at break (autoclave: $100 \pm 60\%$ increase relative to nonsterile, ethanol/UV: $37 \pm 13\%$ increase). Across all metrics, the two sterilization groups performed similarly, with no significant differences found between the two groups.

3.1.3 Phrozen AquaGray 8K

Phrozen AquaGray 8K was also heavily affected by the sterilization process, with statistically significant differences from the nonsterile group for all metrics measured (Fig. 2 Column 3). In tension, the resin became significantly softer and more ductile (Fig. 3 Left). The autoclaved groups saw a decrease in the Young's modulus of $71 \pm 13\%$ ($p < 0.05$, compared to nonsterile), and the ethanol/UV group saw a $65 \pm 19\%$ decrease ($p < 0.05$ compared to nonsterile, $p > 0.05$ compared to autoclaved group). Simultaneously, both groups saw an increase in the elongation at break, a $90 \pm 30\%$ increase

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for the autoclave group relative to nonsterile and a $60 \pm 20\%$ increase for the ethanol/UV group ($p < 0.05$ for both compared to nonsterile, but $p > 0.05$ when compared to each other). This failure happened at a much greater strain (see above) but at a lower Ultimate Tensile Strength, with the autoclave group seeing a decrease of $50 \pm 8\%$ relative to the nonsterile group and the ethanol/UV group seeing a decrease of $55 \pm 12\%$ ($p < 0.05$ compared to nonsterile, $p > 0.05$ compared to each other). Contrary to the decrease in Young's Modulus, both groups saw an increase in Compressive modulus, with a $25 \pm 4\%$ increase relative to the nonsterile group for the autoclave group and a $140 \pm 20\%$ increase for the ethanol/UV group ($p < 0.05$ for both groups compared to nonsterile group). Additionally, unlike the other metrics where no difference existed between the ethanol/UV and autoclave groups, here, the ethanol/UV had a statistically higher compressive modulus than the autoclaved group.

3.2 Elastomeric resins tested are robust to sterilization methods

Using the material testing data from the previously published dataset on candidate resins for bio-hybrid actuators [1], we examined the constitutive models (Fig. 5), Young's Modulus, stress value at 50% strain, and stress value at 100% strain to investigate the effect of sterilization technique on sample mechanics (Fig. 4(a)). These metrics were available for all samples. Additionally, all Asiga Dentagum samples failed during their tensile tests, and from these curves, the Ultimate Tensile Strength and elongation at break were measured (Fig. 4(b)). Only four Formlabs Silicone 40A samples failed across both post-treatment groups and all sterilization groups. Because of this, these failure characteristics were not investigated for the Formlabs Silicone 40A groups.

3.2.1 Formlabs Silicone 40A, IPA and IPA/BuOAc post treatments

Test results for two post-treatment groups were reported with the Formlabs Silicone 40A resin. For Formlabs Silicone 40A post-treated with IPA/BuOAc, no statistically significant differences were observed in any of the metrics measured as a function of sterilization technique (Fig. 4(a), Column 2). Conversely, when the parts were post-treated with only IPA (Fig. 4(a), Column 1), ethanol/UV sterilization led to a statistically higher Stress at 50% and 100% compared to the nonsterile group

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(no difference between autoclave and nonsterile groups). This mean difference was also greater than 5% of the nonsterile mean ($12.2 \pm 1.0\%$ increase relative to nonsterile for Stress at 50% and $13.0 \pm 1.0\%$ increase for Stress at 100%). While it appears as though the manufacturer-recommended post-treatment performed better than the IPA-only treatment (no statistically significant differences in mean values were observed), there was a much larger observed variance in these metrics across all sterilization methods for the recommended post-treatment groups. This variability is also seen in the stress predicted by the constitutive model for strains greater than 50% (Fig. 5(b). This variance is detrimental both to the power of the statistical tests for the given sample size of the study (Tables 1-2) and to the reproducibility of the components produced using this post-treatment method. In addition, while statistically significant differences exist in the ethanol/UV-sterilized group for the IPA-only post-treated group, no difference was seen in the autoclaved group. Autoclaving also led to only a $1.25 \pm 0.13\%$ increase in the Young's modulus relative to the nonsterile group (not significant) and a $1.75 \pm 0.12\%$ increase in the stress at 50% (not significant). Stress at 100% saw a larger increase of $8.4 \pm 0.7\%$ relative to the nonsterile group, but this was also not statistically significant. Thus, for this resin, when it comes to maintaining mechanical properties relative to pre-sterilized conditions and creating reproducible samples, IPA-only post-treatment sterilized in the autoclave performed the best.

This conclusion can only be drawn for metrics within the elastic regime of this material. Because of the force and displacement limitations enforced by the experimental setup, we were unable to test these samples to failure. The max strains (150%-200% or stretches of between 2.5 and 3) and max stresses (1-3 MPa) are less than the manufacturer's provided strain at break of 230% and Ultimate Tensile Strength of 5 MPa, providing a lower bound confirmation of these metrics (Fig. 5(a) and (b)). However, the true failure characteristics could not be tested. That said, given the strains that this material was able to sustain (>150% for all samples in all groups), it is unlikely that this material will fail due to mechanical strain within the operative range of typical biohybrid devices [6], [18].

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3.2.2 Asiga DentaGUM

Autoclave sterilization led to a decrease in the elastic domain properties of DentaGUM resins (Fig. 4 (a), Column 3), with lower Stress at 50% (8.9±0.3%, $p < 0.05$ comparing Autoclave to nonsterile) and Stress at 100% (11.2±0.6%, $p < 0.05$ comparing autoclave to nonsterile). The Young's modulus was also lower (5.1±0.3%), though this did not reach statistical significance (95% confidence interval: $[-0.039, 0.472]$ MPa does not contain 0). Conversely, ethanol/UV sterilization led to no statistically significant difference in Stress at 50% or 100% compared to the nonsterile group, and means were within 5% (stress at 50%: $4.7 \pm 0.3\%$, stress at 100%: $-3.2 \pm 0.2\%$ both relative to the nonsterile group). The Young's modulus was slightly higher than that of the nonsterile group (9.2±0.7%), but this difference was also not statistically significant (95% confidence interval: $[-0.484, 0.242]$ MPa). It was also observed that the nonsterile specimens demonstrated greater strain stiffening. While the Young's modulus (stiffness at 0% strain) was higher for the ethanol group, the stiffness at strains greater than about 40% was greater in the nonsterile group (Fig. 5(c)). The Young's modulus of the ethanol-sterilized group was significantly higher than the autoclaved-sterilized group (95% confidence interval: $[-0.551, -0.089]$ MPa). Despite these changes in the elastic regime of these materials, there were no statistically significant differences in the failure properties of the different sterilization groups, with both the autoclave- and ethanol/UV-sterilized group means within 5% of the nonsterile mean for Ultimate Tensile Strength (Fig. 4(b)). When comparing the elongation at break, the mean for the autoclaved group was slightly greater than 5% of the nonsterile mean (6.5±0.6%), but the variability in this metric for both the autoclaved and nonsterile groups prevents this from being a significant difference. Given these results, ethanol/UV sterilization may be the more appropriate sterilization technique for this resin to preserve pre-sterilization properties.

4 Discussion

The analysis presented here highlights the importance of understanding the mechanical properties of 3D printed parts when exposed to the environments and sterilization conditions in which they will be used. The rigid resin groups were, in general, more greatly affected by the different steriliza-

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tion techniques, with many of the soft resins showing no significant difference in their mechanical properties. Both Phrozen AquaGray 8K and Liqcreate Bio-Med Clear became much less stiff in tension after sterilization, and Phrozen AquaGray 8K became much more ductile. Asiga DentaGUIDE was more stable in its properties, with only the compressive modulus being significantly different. Of particular importance is that the mechanical properties of many of the rigid resins varied significantly from the manufacturer-reported values when tested with the prolonged exposure to cell culture conditions used to generate the dataset analyzed here [1]. Similar to the recommendations by [28] on the tensile testing of PLA, this analysis further demonstrates the critical need for the design of biomedical devices and biohybrid robots to account for use in cell culture conditions. Finally, while the mean properties before and after sterilization were not very different for the two Formlabs Silicone post-treatment groups, the IPA/BuOAc post-treated samples saw a much larger property variability compared to the IPA-only group, suggesting that only IPA should be used for post-treating these parts.

4.1 Causes of environmental degradation:

4.1.1 Interactions between saline exposure and sterilization

The effects of the sterilization techniques on mechanical properties may have also been exacerbated by the PBS incubation. As reported in the data description paper that was the source of data for the analysis presented here [1], the stiffness of the soaked but nonsterile samples was much lower than the values reported by the company for both Liqcreate Bio-Med Clear and Phrozen AquaGray 8K (with insufficient data to determine the same for Asiga DentaGUIDE). Follow-up experiments with soaked and non-soaked beams, as presented in that report, aligned with the interpretation that the difference is caused by the soak itself. The impact of saline exposure on plastic materials has been observed previously [35], [36], [37], where it was also observed that saline exposure weakened parts (and did so more than exposure to distilled water). Rege and Lakkad [35] suggest that moisture absorption by the materials may be causing this, as water diffusing into the polymer

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matrix could cause internal stresses and damage within the matrix. For saline samples, there was the additional diffusion of ions, which were preferentially absorbed by the samples. This effect is likely at play within our samples as well, as the unsterilized samples were much softer than the samples never exposed to saline. The methods of sterilization may have exacerbated this effect as well, causing the softening to be worse in the sterilized groups, as the sterilization techniques may have mildly weakened the matrix of the materials. If, prior to soaking in saline, the matrix was partially damaged, then water and ions may more readily diffuse into the samples, causing more damage than in the nonsterile samples.

The softening is far more pronounced in Phrozen AquaGray than in the two resins developed for medical applications. If these other resins were designed to better resist degradation from sterilization, then one would expect to see less additional saline absorption after sterilization if sterilization is, in fact, causing micro-defects. To test this directly, the mass of samples could be measured at each stage of the post-processing to determine the levels of saline absorption in each group. Additionally, to isolate just the effects of sterilization in the absence of saline, one could repeat this study without the saline incubation. These tests would allow one to determine if saline-induced and sterilization-induced softening effects are related and how they interact. Finally, some methacrylate-based plastics are plasticized by water absorption [37]. To determine if any of these softening effects are due to the glass transition temperature being lowered below 37°C (at which temperature the materials were tested), differential scanning calorimetry (DSC) and/or dynamic mechanical thermal analysis (DMTA) could be conducted on samples before and after saline exposure.

The effects of saline exposure and its interaction with sterilization may be less pronounced in the soft resin samples because of their inherent hydrophobicity, minimizing the amount of water that is absorbed. In addition to weighing samples, one could test the hydrophobicity of the different materials by, e.g., a contact angle measurement [38] to determine if there is a relationship between a) hydrophobicity, b) amount of absorbed water, and c) changes in material properties. If increasing the hydrophobicity of a material is protective of its mechanical properties by minimizing water absorption, this could be a key design consideration when either developing new materials

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for saline contact or developing post-processing steps, which could include a surface coating step. Alternatively, if the softening effects in the rigid resins are caused by plasticization, this may also be non-obvious in the elastomer resins, as they are already past their glass transition under normal conditions.

4.1.2 Anisotropic effects of saline and sterilization

For the rigid resin samples, both before and after sterilization, the compressive modulus was systematically lower than the tensile modulus (Fig. 2). For a homogeneous and isotropic material, this would be unexpected. Additionally, the sterilization induced different changes in the two types of samples, with the tensile samples becoming softer and the compressive samples stiffer. Both of these effects may be understood as the result of printing-induced anisotropy [39], [40], [41]. The tensile dog bone samples were printed flat against the build plate such that the build layers were parallel to the direction of force application during tests. The compressive samples, conversely, were printed such that the build layers were perpendicular to the direction of force application. Thus, for the dog bone samples, the build planes and the inter-layer bonds act in parallel to each other, and in the compressive samples, they act in series. If the build layers were stiffer than the inter-layer bonds, then this would manifest in the tensile samples being stiffer than the compressive samples. This is because the parallel arrangement in the tensile samples will be dominated by the stiffer element, whereas the series arrangement in the compressive samples would be more affected by the softer element. This printing-induced anisotropy could be investigated by testing dog bone samples printed at different elevations relative to the build plate so that the build planes sit at different angles to the test direction [39].

This still leaves the observation that tensile samples became softer with sterilization, while compressive samples became stiffer. This could be related to the dual function of water in these samples. As the water diffuses into the build layers of the samples, it would damage the polymer matrix, tending to decrease the stiffness of the matrix. However, if it becomes trapped between layers or fills voids in the material (potentially induced by sterilization), it would serve as an incompressible hydraulic element. The effect on stiffness from this element would be related to the

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volume fraction of the fluid in the sample [42]. This hydraulic element could lead to a stiffening of the material, particularly in the softer inter-layer bonds. Thus, these two effects would compete with each other to change the material properties. How they will differentially affect the two sample types can be explored by considering how the water will diffuse through the samples.

Because of the aspect ratio of the tensile dog bones, this diffusion would occur predominantly through the flat faces of the sample (i.e., through the build planes). Therefore, the elements that are dominating the stiffness of the tensile samples would be weakened, leading to the lower modulus that we observed. For the compressive samples, on the other hand, diffusion would likely occur predominantly radially through the cylindrical wall (i.e., parallel to the build planes). This may lead to less damage to the build layers while saturating the inter-layer regions, leading to a more pronounced hydraulic effect. This would also be amplified by the greater cross-sectional area of the cylindrical elements. As discussed previously, the diffusion of water may be amplified by sterilization causing microdamage to the matrix. Therefore, if anisotropic effects of the water penetration are occurring, we would expect those effects to be amplified by the sterilization (e.g., the tensile samples would get even softer, and the compressive samples would get stiffer). There are likely combinations of these effects happening in both samples, and a wider range of conditions would need to be factorially addressed to confidently claim how these effects are interacting.

4.2 Limitations:

The dataset analyzed for this study has a few limitations for analysis. In particular, the resins were only tested in single-stroke experiments. While this follows the current ASTM standard protocols, these experiments only provided limited information about the performance characteristics of these materials, particularly for the soft resins. For a more thorough characterization of these materials, cyclic testing should be performed for tensile and compression loading conditions. Many elastomeric materials exhibit the Mullin's effect [43], and these single-stroke experiments only allow for the investigation of the virgin stress-strain curve. Using cyclic tests that span the range of strains the material is expected to experience *in situ*, a more performance-relevant analysis could be conducted. Additionally, these synthetic materials have roles as supportive substrates and skeletons in biohybrid actuators, which may undergo repetitive loading cycles for multiple functions, such as locomotion

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[9], [10], [11], [12], [13], [14] or grasping [9], [18], which further highlight the need for future cyclic testing. As the goal of this study was primarily to analyze the existing published dataset for the effect of sterilization techniques on mechanical properties, these factors were excluded. However, for a more complete analysis of the mechanical properties of these resins, these variables should be investigated in future experimental studies.

These cyclic tests would also allow for the investigation of another important factor that was not investigated here – viscoelasticity. Many of these elastomeric resins also exhibit some degree of viscoelasticity, and this could be characterized using cyclic tests at varying strain rates. This strain rate dependence was not confined only to the soft resins but was also demonstrated here by the rigid resins. The tensile samples were tested at different strain rates than the compression samples in accordance with the ASTM standards (∼70%/min for tensile tests and ∼25%/min for compression tests. Exact strain rates are dependent on the rest geometry, as a fixed displacement rate was used for the two protocols). When we calculated moduli for these different loading conditions, we saw much lower compressive moduli than tensile moduli for all resins and sterilization methods. This would be expected for viscoelastic materials tested at different strain rates, as the Young's modulus tends to increase with strain rate [44]. In future experiments, these values should either be measured at various strain rates to capture their strain rate dependence or a single strain rate should be chosen that aligns with the expected operational strain rates for the material.

4.3 Conclusions:

Based on the analysis reported here, many of the rigid resins were significantly impacted by the sterilization procedures common in biomedical research labs. For the tested resins, Asiga DentaGUIDE or Formlabs Silicone 40A are the leading potential candidates for printing biohybrid base components using low-cost 3D-printing hardware, as their mechanical properties were minimally impacted by sterilization. Together, these findings provide a basis for researchers to select candidate materials for use in biohybrid robotics with low-cost lab-based manufacturing approaches, with a goal of broadening access to biohybrid robotics research.

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5 Acknowledgements

During the preparation of this work, the authors used the Grammarly Chrome Extension in order to improve grammar, spelling, and clarity. No generative AI tools were used to create new text. Any changes and suggestions made by the AI editing tool were subsequently reviewed by the authors, and the authors take full responsibility for the content of the publication. Fig. 1 was generated using tools and assets from Biorender.

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7 Conflicts of interest

KD is the founder of Mimetic Ink Corp., a company developing biocompatible 3D printing capabilities for laboratory research. No products from Mimetic Ink Corp. were used in this study.

8 Data availability

All data to reproduce the work presented in this paper are described in Liao *et al.* 2025 [1] and are available on GitHub (https://github.com/CMU-BORG/Commercial-Resins-for-Biohybrids-SciData). An archived version of this dataset is available on Zenodo (doi: 10.5281/zenodo.15801032). Code was modified from this original publication to produce the plots and conduct statistical analysis in this work. All code to replicate plot and statistical analysis is available on GitHub (https://github.com/CMU-BORG/Effects-on-Resin-Mechanical-Properties). Any data from Liao *et al.* 2025 required for these plots is included in the new repository for completeness. An archived version of this repository is available on Zenodo (doi: 10.5281/zenodo.17354784).

9 Author contribution

Conceptualization: ASL, KD, ABI, VAW **Methodology**: ASL, KD, ABI, MJB, VAW **Software**: ASL, MJB **Validation**: ASL, KD, ABI, MJB, VAW **Formal Analysis**: ASL, MJB, VAW **Investigation**: ASL, KD, ABI, AS, SS, BC, JMS, RA, MJB **Resources**: ASL, KD, VAW **Data Curation**: ASL, ABI, MJB, VAW **Writing - Original Draft**: ASL, MJB, VAW **Writing - Review & Editing**: ASL, KD, ABI, AS, MJB, SS, BC, JMS, RA, YJZ, VAW **Visualization**: ASL, MJB, VAW **Supervision**: ASL, YJZ, VAW **Project Administration**: ASL, VAW **Funding Acquisition**: ASL, KD, ABI, MJB, SS, YJZ, VAW

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Table Captions

Table 1: Average geometries of mechanical testing tensile samples. The gauge length is defined by the distance between the jaws that secured the sample in the test setup prior to loading. N is the sample size. For the original CAD geometries used to print the samples, refer to [1].

Table 2: Average geometries of mechanical testing compression samples. The gauge length is defined by the distance between the jaws that secured the sample in the test setup prior to loading. N is the sample size. For the original CAD geometries used to print the samples, refer to [1].

Figure Captions

Figure 1: Analysis Overview. Samples were fabricated using the Phrozen Sonic Mini 8K 3D printer. After post-processing, the prints were sterilized using either an autoclave or 70% ethanol with UV exposure. Samples were then incubated for 16 hours at 37°C in 1x PBS to mimic cell culturing conditions. Finally, samples underwent either tensile or compressive mechanical test, from which constitutive properties and failure characteristics were measured for each resin and sterilization method. Created in BioRender. Webster-Wood, V. (2025) https://BioRender.com/s0z8jb9

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Figure 3: Rigid Resin Model Comparisons. Left: Uniaxial tensile properties; right: uniaxial compressive properties. Top: Asiga DentaGUIDE; Middle: Liqcreate Bio-Med Clear; Bottom: Phrozen AquaGray 8K. Main plots show the mean Hookean model response for each sterilization condition (shaded region shows ±1 standard deviation). Subplots show the stress and strain data for each sample along with the group mean model fit. Each sample's data has been corrected (according to ASTM D638) such that the fitted Hookean model for that dataset passes through the origin. For tensile properties, curves end at sample failure, but for compressive samples, curves end when reaching machine limits. All vertical axes are consistent across tensile and compressive plots, respectively.

Figure 4: Elastomeric Resin Material Properties. (a) Elastic domain properties. The Young's Modulus [MPa] (top row, calculated from Yeoh model parameters as $E = 6C_1$), Stress at 50% strain (middle row), and Stress at 100% strain (bottom row) are reported for the three elastomeric resin groups and for each sterilization technique (NS - nonsterile, autoclave, EtOH - ethanol/UV exposure). (b) Failure properties for Asiga DentaGUM (top: Ultimate Tensile Strength, bottom: Elongation at Break). Properties are not reported for Formlabs Silicone because no samples broke within the machine limits. Bar height indicates the mean value, and error bars show ± standard deviation. Marks show individual data points, with one data point per specimen tested. Individual data points are not shown for the Young's modulus data because these distributions were computed using a bootstrapping approach. Note that the scales for Asiga DentaGUM in (a) are higher than the rest of the charts. Otherwise, scales are consistent across each row. Horizontal dashed lines indicate ±5% of the corresponding nonsterile group mean. Solid lines above bars indicate statistical difference in means at the 5% level.

Figure 5: Elastomeric Resin Model Comparisons for Formlabs Silicone 40A with (a) IPA only and (b) IPA/BuOAc post-treatment, and (c) Asiga DentaGUM. Each curve shows the mean Yeoh model response for each sterilization condition (NS - nonsterile, autoclave, EtOH - ethanol/UV exposure). Shaded region shows ±1 standard deviation. For each sterilization technique, the coefficient of determination (R^2) is calculated for the mean

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model response against the full dataset and is reported in the associated legend.

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Tables

Table 1: Average geometries of mechanical testing tensile samples. The gauge length is defined by the distance between the jaws that secured the sample in the test setup prior to loading. N is the sample size. For the original CAD geometries used to print the samples, refer to [1].

Sterilization & Resin			N	Gauge Length	Thickness	Width
Autoclave	*Rigid*	DentaGUIDE	7	6.23 ± 1.27	0.71 ± 0.03	1.06 ± 0.07
		AquaGray 8K	6	7.44 ± 0.34	0.72 ± 0.07	1.12 ± 0.04
		Bio-Med Clear	6	5.51 ± 0.64	0.77 ± 0.04	1.10 ± 0.07
	Elastomeric	DentaGUM	6	26.23 ± 1.00	1.51 ± 0.03	3.02 ± 0.04
		Silicone 40A IPA	6	28.25 ± 1.27	1.51 ± 0.03	2.94 ± 0.04
		Silicone 40A IPA/BuOAc	6	26.34 ± 1.06	1.21 ± 0.04	2.86 ± 0.04
Ethanol/UV	*Rigid*	DentaGUIDE	6	7.59 ± 0.54	0.72 ± 0.02	1.00 ± 0.05
		AquaGray 8K	6	7.01 ± 0.41	0.74 ± 0.09	1.11 ± 0.02
		Bio-Med Clear	6	7.32 ± 0.40	0.73 ± 0.06	1.14 ± 0.08
	Elastomeric	DentaGUM	6	25.57 ± 0.65	1.50 ± 0.04	3.00 ± 0.01
		Silicone 40A IPA	6	28.30 ± 1.10	1.53 ± 0.04	2.89 ± 0.06
		Silicone 40A IPA/BuOAc	6	25.91 ± 0.59	1.21 ± 0.04	2.87 ± 0.03
Nonsterile	*Rigid*	DentaGUIDE	4	7.02 ± 0.23	0.69 ± 0.03	0.98 ± 0.03
		AquaGray 8K	5	7.10 ± 0.58	0.72 ± 0.05	1.09 ± 0.05
		Bio-Med Clear	5	7.23 ± 0.37	0.73 ± 0.07	1.05 ± 0.04
	Elastomeric	DentaGUM	6	26.29 ± 0.75	1.50 ± 0.04	3.03 ± 0.02
		Silicone 40A IPA	6	27.02 ± 1.02	1.57 ± 0.02	2.90 ± 0.04
		Silicone 40A IPA/BuOAc	6	25.57 ± 0.66	1.21 ± 0.04	2.87 ± 0.06

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Table 2: Average geometries of mechanical testing compression samples. The gauge length is defined by the distance between the jaws that secured the sample in the test setup prior to loading. N is the sample size. For the original CAD geometries used to print the samples, refer to [1].

Sterilization & Resin			N	Gauge Length	Diameter
Autoclave	*Rigid*	DentaGUIDE	6	4.99 ± 0.06	2.47 ± 0.05
		AquaGray 8K	6	5.17 ± 0.04	2.49 ± 0.03
		Bio-Med Clear	6	5.03 ± 0.01	2.49 ± 0.02
	Elastomeric	DentaGUM	6	5.03 ± 0.03	11.42 ± 0.03
		Silicone 40A IPA	6	5.08 ± 0.05	11.51 ± 0.06
		Silicone 40A IPA/BuOAc	6	4.71 ± 0.04	11.30 ± 0.04
Ethanol/UV	*Rigid*	DentaGUIDE	6	5.02 ± 0.02	2.47 ± 0.02
		AquaGray 8K	6	5.13 ± 0.05	2.49 ± 0.02
		Bio-Med Clear	6	5.06 ± 0.02	2.48 ± 0.03
	Elastomeric	DentaGUM	6	5.00 ± 0.01	11.34 ± 0.06
		Silicone 40A IPA	6	5.04 ± 0.04	11.39 ± 0.13
		Silicone 40A IPA/BuOAc	6	4.70 ± 0.05	11.29 ± 0.02
Nonsterile	*Rigid*	DentaGUIDE	6	5.01 ± 0.03	2.47 ± 0.03
		AquaGray 8K	7	5.15 ± 0.08	2.48 ± 0.02
		Bio-Med Clear	6	5.07 ± 0.12	2.60 ± 0.12
	Elastomeric	DentaGUM	6	5.01 ± 0.02	11.36 ± 0.02
		Silicone 40A IPA	6	5.01 ± 0.01	11.46 ± 0.04
		Silicone 40A IPA/BuOAc	6	4.67 ± 0.04	11.33 ± 0.04

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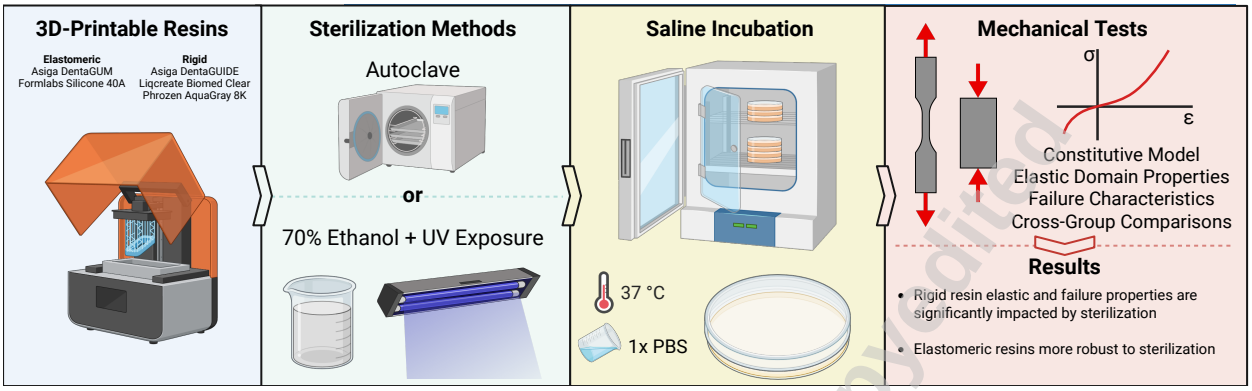
Figures



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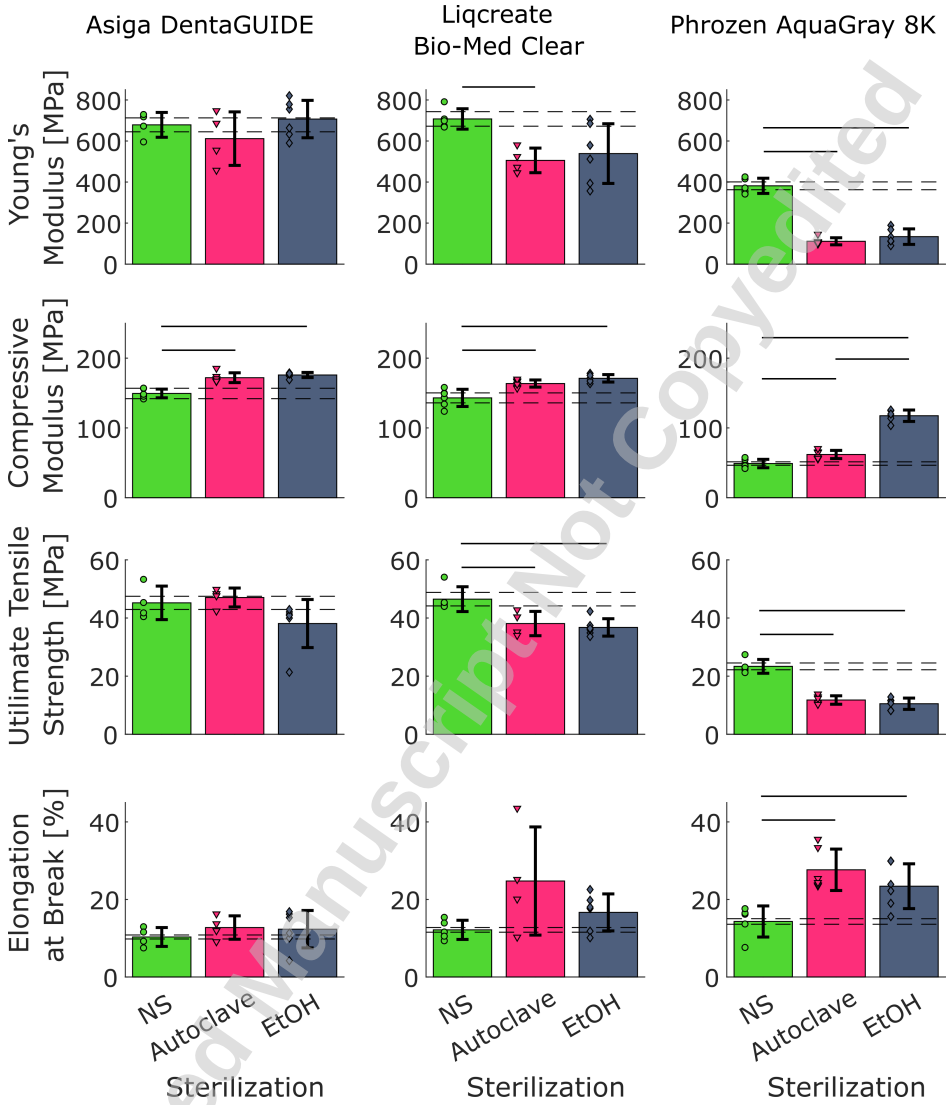

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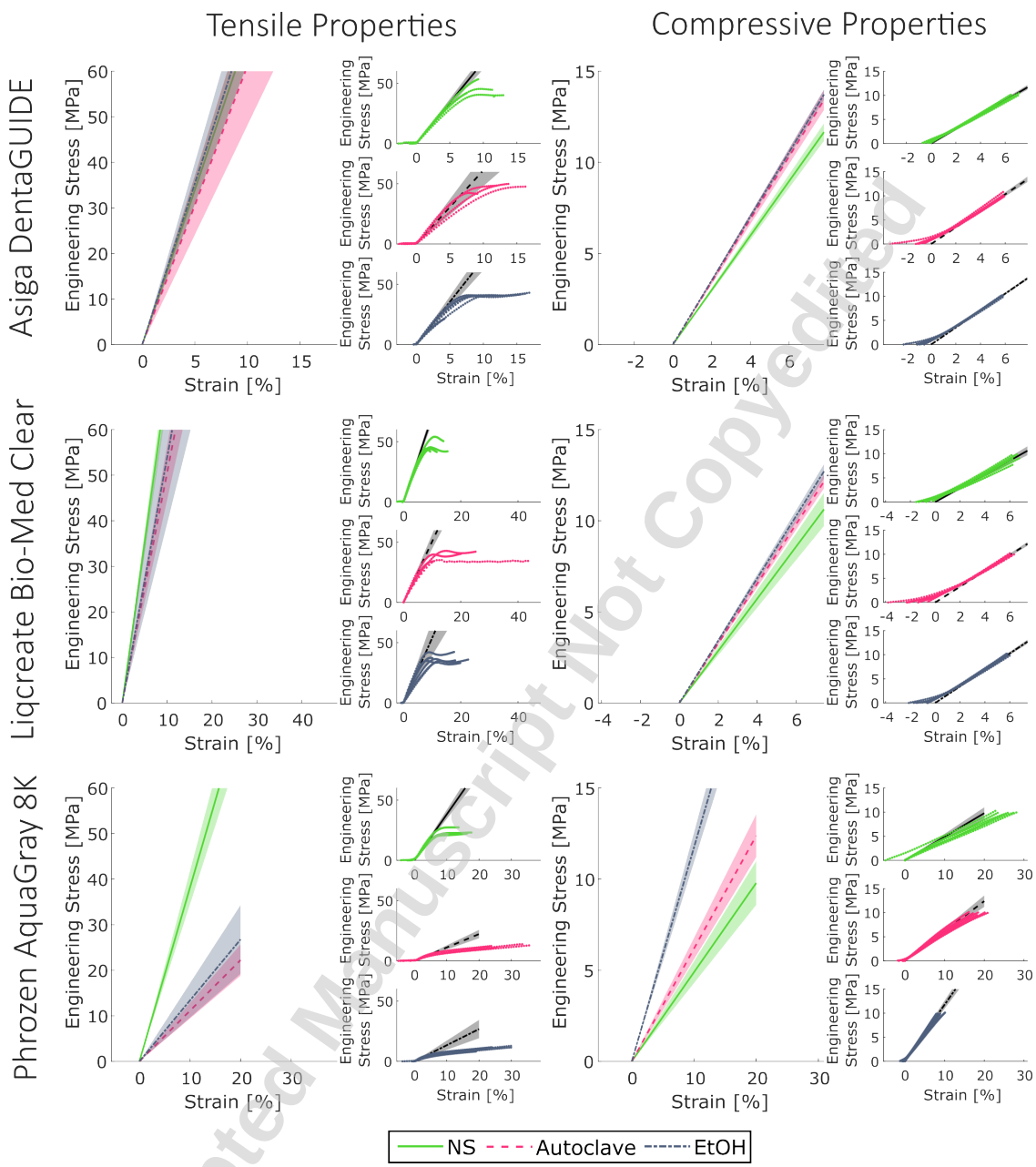

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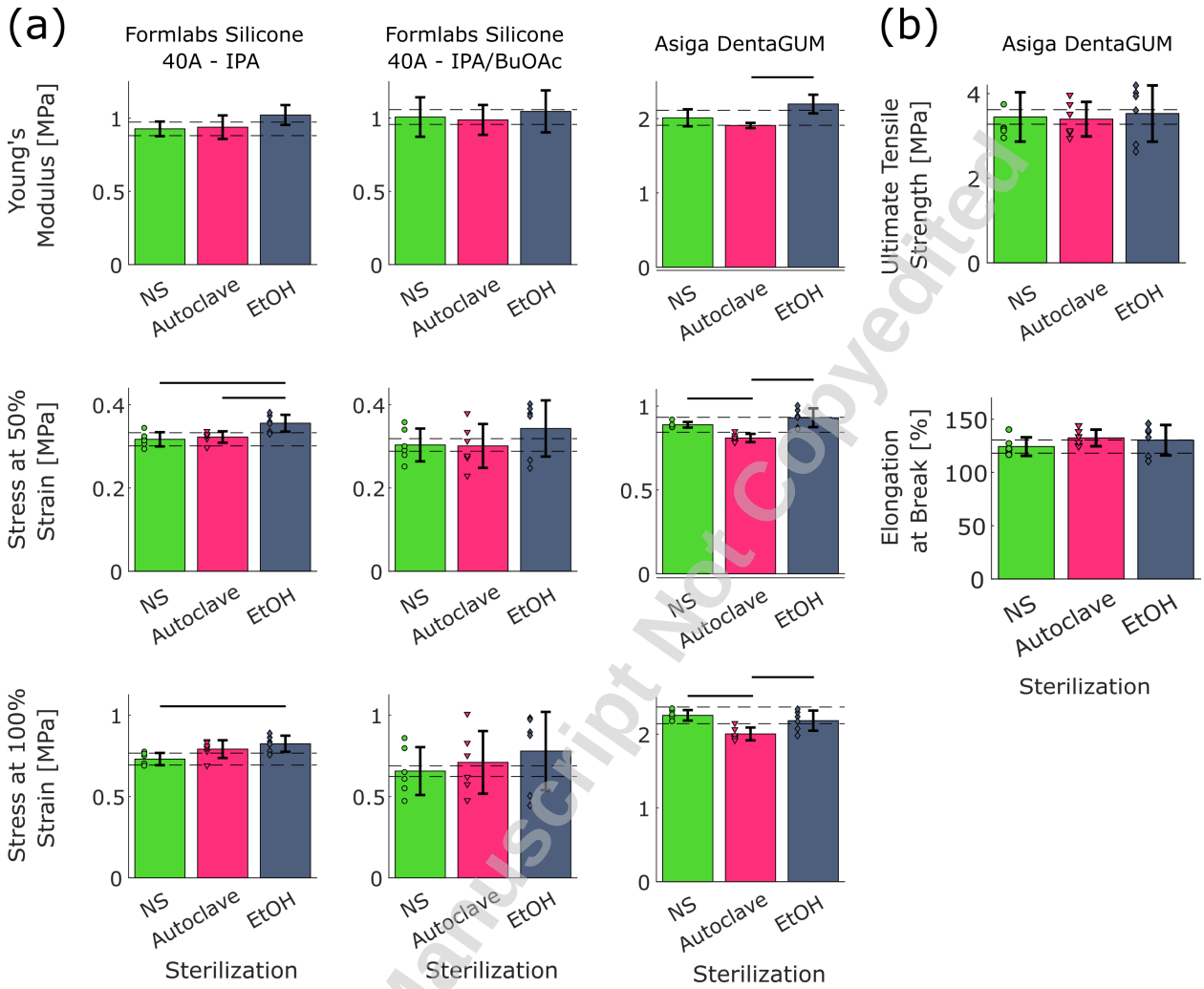

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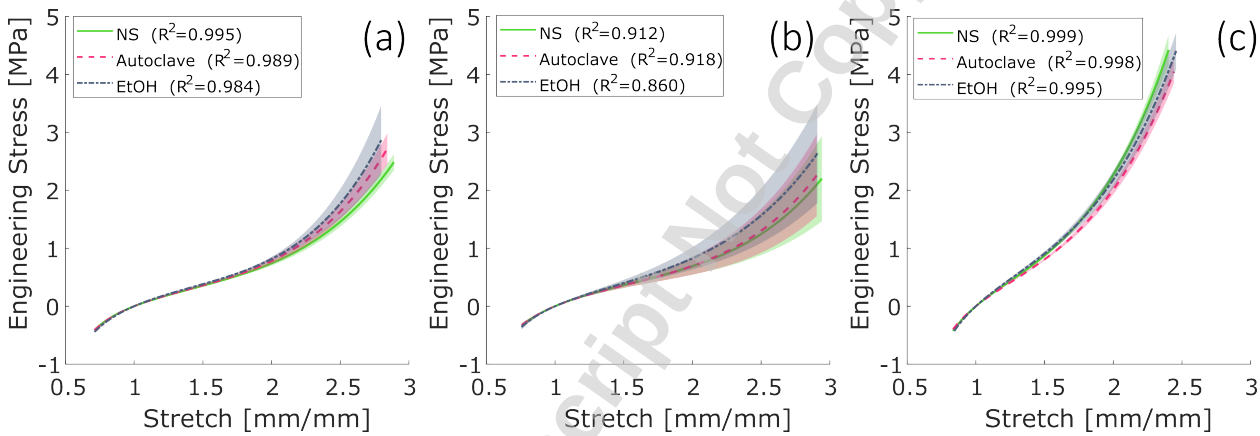

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